

Two-term asymptotic expansion of the chart standard integral

Claude, with Daniel Murfet

2026-09-06

Abstract

The O -form of the second-order result: for $\xi(u) = \xi_0 + cu^m$,

$$Z_{\text{chart}}(n) = \frac{1}{2k} S_{\mu_1}(\xi_0) n^{-\mu_1} + \beta c \frac{1}{2k} S_{\mu_2+1/2}(\xi_0) n^{-\mu_2} + O(n^{-\mu_3}) \quad (n \rightarrow \infty),$$

with $\mu_1, \mu_2, \mu_3 = (h+1)/(2k), (h+m+1)/(2k), (h+2m+1)/(2k)$ three consecutive points of $\Lambda(h, k)$. The exponentially small boundary tails — including the $\sqrt{n} e^{-\varepsilon n}$ one from the first-order insertion — are absorbed into the $O(n^{-\mu_3})$. Zero `sorry/axiom`.

1 Setting

Unit 28 bounds the two-term error by $C_2 n^{-\mu_3} + \text{tail}_0 + |\beta c| \text{tail}_1$ for $n \geq 1$, with the tails explicit integrals. To read this as an asymptotic expansion in the paper's sense one needs the tails to be $O(n^{-\mu_3})$: $\text{tail}_0 \leq K_0 e^{-\varepsilon n}$ directly from unit 25, while $\text{tail}_1 = \sqrt{n} \int_b^\infty u^{h+m+k} e^{\text{pop}} \leq \sqrt{n} K_1 e^{-\varepsilon n}$, and both $e^{-\varepsilon n}$ and $\sqrt{n} e^{-\varepsilon n}$ are $o(n^{-\mu_3})$.

2 The thing formalised

```
theorem standardIntegral1D_second_order_isBig0 ( c b : ) (h k m : )
  (h : 0 < ) (hb : 0 < b) (hk : 0 < k) :
  (fun n : => Z_chart n
    - (1/(2k)) n^(- ) S_{ }( ) - c ((1/(2k)) n^(- ) S_{ +1/2}( )))
    =0[atTop] fun n : => n ^ (- ))
```

3 Proof strategy

`IsBig0.of_bound` with the constant $C_2 + K_0 + |\beta c| K_1$, eventually for $n \geq 1$. The two little- o facts are `isLittle0_exp_neg_mul_rpow_atTop` at exponent $-\mu_3$, and for the $\sqrt{n}$ -weighted one, `IsBig0.mul_isLittle0` of $\sqrt{n} = O(\sqrt{n})$ with the same lemma at $-\mu_3 - \frac{1}{2}$, followed by `IsLittle0.congr'` using $\sqrt{n} \cdot n^{-\mu_3-1/2} = n^{-\mu_3}$ for $n > 0$ (`Real.sqrt_eq_rpow`, `rpow_add`). Their `.bound one_pos` instances are pulled into the same `filter_upwards` as $n \geq 1$; the rest is the unit-28 inequality with the tails replaced by $K_i e^{-\varepsilon n}$ (`standardIntegral1D_tail_le` at h and at $h + m + k$) and then by $K_i n^{-\mu_3}$, finished by `gcongr` and `ring`.

4 Roadblocks and resolutions

None: first-try compile. Idioms worth noting — `.bound one_pos` produces norms of un-beta-reduced lambda applications, so convert with `simp only [Real.norm_eq_abs]` (which beta-reduces) before `rw [abs_of_pos ...]`; the first-order tail integrand is rewritten to $\sqrt{n} \cdot u^{h+m+k} e^{\text{pop}}$

by `funext + pow_add` so that the plain tail lemma applies; and `set` at the top folds the goal but not later-derived hypotheses, so `rw [← h ]` at `hmain` re-aligns them.

5 What was Mathlib, what was new

Mathlib: `IsBig0.of_bound`, `IsBig0.mul_isLittle0`, `IsLittle0.congr'`, `IsLittle0.bound`, `isLittle0_exp_neg_mul_rpow_atTop`, `Real.Gamma_pos_of_pos`. New: the two-term asymptotic expansion of the chart standard integral, the first formal instance of `cor:standardintegralexp` beyond leading order.

6 Lessons

Keep the explicit-inequality theorem (unit 28) and the *O*-packaging (this unit) separate: the former is what one computes with (constants visible), the latter is what one quotes; the packaging step is short once the little-*o* scales are named.

7 Follow-ups

Higher-order terms follow the same pattern from `standardIntegral1D_monomial_expansion` once the remainder $\sum_{p \geq N}$ is bounded; the multivariate tree remains the substantive gap.